

AEON Trading Bot

Complete System Documentation

Live URL: <https://aeon-paper-trade.preview.emergentagent.com>

1. Executive Summary

AEON is an autonomous cryptocurrency paper trading bot that operates 24/7, analyzing markets and executing trades based on technical analysis, smart money concepts, and derivatives data.

Property	Value
Bot Type	Autonomous Paper Trading Bot
Markets	Crypto Futures (25 pairs)
Exchanges	MEXC, Binance (data feeds)
Trading Mode	Paper Trading (simulated)
Operation	24/7 Autonomous
AI Powered	Yes - GPT-4o for conversation

Current Statistics

- **Total Trades:** 103 +
 - **Win Rate:** 19%
 - **Active Positions:** 2-3
 - **Trading Pairs:** 25
 - **Active Users:** 38 +
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2. System Architecture

Technology Stack

Backend:

- Framework: FastAPI (Python 3.11)
- Database: MongoDB with Motor (async driver)
- Task Scheduling: APScheduler
- Async Runtime: asyncio
- Process Manager: Supervisor
- Web Server: Uvicorn

Frontend:

- Framework: React 18

- Build Tool: Create React App + Craco
- UI Components: Shadcn/UI
- Styling: Tailwind CSS
- Charts: Recharts
- Icons: Lucide React

Infrastructure:

- Hosting: Emergent Platform (Kubernetes)
 - Container: Docker
 - Proxy: Nginx
 - SSL: Cloudflare
 - Backend Port: 8001
 - Frontend Port: 3000
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3. AI & API Integrations

AI Services

OpenAI GPT-4o (Conversation AI)

- **Purpose:** Powers "Talk to Aeon" voice/text chat
- **Model:** gpt-4o-mini
- **Integration:** Emergent LLM Key (Universal Key)
- **Features:**
 - Natural language trading questions
 - Market analysis explanations
 - Personalized responses
 - Context-aware conversation

OpenAI Whisper (Speech-to-Text)

- **Purpose:** Transcribe user voice input
- **Integration:** Emergent LLM Key
- **Endpoint:** /api/voice/transcribe

Edge TTS (Text-to-Speech)

- **Purpose:** Generate Aeon's voice responses
- **Voices:** Multiple options (male, female)
- **Output:** MP3 audio

Market Data APIs

API	Purpose	Rate Limit
MEXC Exchange	Primary price data & orderbook	1200/min
Binance	Technical indicators, funding rates	1200/min
CoinGecko	Market cap, global data	10-30/min

CoinGlass	Derivatives data, liquidations	30/min
CryptoCompare	News aggregation	-
CryptoPanic	Crypto news feed	-

Telegram Bot API

- **Token:** TELEGRAM_TOKEN (in .env)
- **Webhook:** Auto-configured on startup
- **Endpoint:** /webhook (receives updates)
- **Features:** Inline keyboards, Markdown formatting, Real-time alerts

4. Trading Engine Specifications

Confluence Scoring System (Weighted)

Category	Weight	Indicators
Core Technicals	60%	RSI, MACD, Bollinger Bands, EMAs (9/21/50), Stochastic
Market Structure/SMC	20%	HH/HL, LL/LH, BOS, CHoCH, FVG, Order Blocks
Derivatives/Order Flow	20%	Funding Rate, Open Interest, L/S Ratio, CVD

Entry Requirements

- Minimum 4/5 confluences agreeing
- Minimum 80% confidence score
- R:R ratio minimum 2:1
- ATR < 2x average (no choppy markets)
- Pair not blacklisted or on cooldown

Trade Styles (Dynamically Assigned)

Style	Timeframes	Leverage	ATR Trigger
SCALP	5m, 15m	50-200x	ATR > 2% (high vol)
DAY	1h, 4h	20-75x	ATR 1-2% (medium)
SWING	4h, 1d	10-25x	ATR < 1% (low vol)

Position Sizing

Kelly Criterion Formula: $f = (p \times b - q) / b$

Parameter	Value
Minimum Position	\$500
Maximum Position	\$2,500
Base Position	\$1,000
Risk Per Trade	1% max

Position Scaling (50/50 Entry):

- Initial Entry: 50% of calculated size
- Scale-In: Remaining 50% when price moves 0.3-1.5% in favor

Risk Management

Feature	Description
Auto-Blacklist	Pairs with < 30% win rate after 10 trades
Cooldown Period	4 hours after losing trade on same pair
Volatility Filter	Skip if ATR > 2x 20-period average
Max Open Trades	15 positions maximum
Min Confidence	80% threshold for trade entry
Trailing Stop	Moves to breakeven at +1% profit
Momentum Exit	Early exit if RSI diverges while in profit

Supported Trading Pairs (25)

BTC/USDT, ETH/USDT, SOL/USDT, BNB/USDT, XRP/USDT, DOGE/USDT, ADA/USDT, AVAX/USDT, LINK/USDT, DOT/USDT, ATOM/USDT, UNI/USDT, LTC/USDT, ARB/USDT, OP/USDT, INJ/USDT, NEAR/USDT, APT/USDT, FIL/USDT, TRX/USDT, POL/USDT, SHIB/USDT, BCH/USDT, ETC/USDT, XLM/USDT

5. Technical Indicators

Core Technicals

Indicator	Configuration	Usage
RSI	Period: 14	< 30 = oversold (long), > 70 = overbought (short)
MACD	12,26,9	Crossovers + histogram expansion
Bollinger Bands	20,2	Band touches + squeeze detection
EMAs	9/21/50	Stack alignment, pullback entries
Stochastic	14,3,3	Overbought/oversold crossovers
ATR	Period: 14	Stop loss calculation, volatility filtering

Smart Money Concepts (SMC)

- Break of Structure (BOS)
- Change of Character (CHoCH)
- Fair Value Gaps (FVG)
- Order Blocks
- Liquidity Sweeps
- Premium/Discount Zones

Derivatives Data

- Funding Rate (from Binance/MEXC)

- Open Interest changes
 - Long/Short Ratio
 - Cumulative Volume Delta (CVD)
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6. Frontend Dashboard

Navigation Menu

Page	Description
Dashboard	Main stats, positions, market data
Trading	Position management, trade details
History	Closed trades history
Alerts	Price alerts management
Backtest	Strategy backtesting
Intel	Market intelligence
Analytics	Performance charts
SMC	Smart Money analysis
Journal	Trade journaling
Commands	Telegram command reference
Health	System health monitoring
Settings	Configuration

Dashboard Components

1. **Stats Bar** - Status, Win Rate, PnL, Open Trades, Leverage, Users
2. **Daily Goal Progress** - Visual progress toward daily target
3. **Live Positions** - Real-time position display with PnL
4. **Performance Stats** - Best/Worst pairs, Risk Management
5. **Live Market Data** - BTC, ETH, SOL prices with buy/sell imbalance

Position Modal Features

- PnL (percentage and USD)
 - ROI Badge
 - Risk Assessment
 - Margin Ratio
 - Entry/Current/Target/Stop prices
 - Quick Close buttons (25%, 50%, 75%, 100%)
 - Trailing Stop setter
 - Trade confirmations list
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7. Telegram Bot Commands (50 +)

Trading Commands

/trade [COIN] - Quick trade analysis
/scan [COIN] - Full market scan with SMC
/positions - View all open positions
/close [COIN] - Close specific position
/closeall - Emergency close all positions
/pnl - Today's PnL summary

Analysis Commands

/ta [COIN] - Technical analysis (RSI, MACD, BB)
/smc [COIN] - Smart Money Concepts analysis
/structure [COIN] - Market structure (HH/HL, BOS)
/divergence [COIN] - RSI divergence detection
/funding - Funding rates
/fear - Fear & Greed index

Stats Commands

/stats - Trading statistics
/accuracy - Alert accuracy stats
/leaderboard - Coin performance ranking
/riskcheck - Risk assessment

8. API Endpoints (70+)

Trading Endpoints

Method	Endpoint	Description
GET	/api/trading/v2/stats	Full trading statistics
GET	/api/trading/v2/live-positions	Live positions with PnL
GET	/api/trading/v2/closed	Closed trades history
POST	/api/trading/v2/close/{symbol}	Close position
POST	/api/trading/v2/trail/{symbol}	Set trailing stop
POST	/api/trading/v2/toggle-scaling	Toggle position scaling

Dashboard & Stats

Method	Endpoint	Description
GET	/api/stats/dashboard	Best/worst pairs, blacklist, cooldowns
GET	/api/accuracy/leaderboard	Coin performance leaderboard
GET	/api/system/health	Service health status

Analysis

Method	Endpoint	Description
GET	/api/analysis/{symbol}	Technical analysis

GET	/api/smc/{symbol}	Smart Money Concepts
GET	/api/derivatives/{symbol}	Derivatives data
GET	/api/sentiment/composite	Composite sentiment

Voice

Method	Endpoint	Description
POST	/api/voice/respond	Voice conversation (TTS)
POST	/api/voice/transcribe	Transcribe audio (STT)

9. Database Schema (MongoDB)

Collections

v2_open_trades

```
{
  "_id": "trade_103",
  "symbol": "BCH/USDT",
  "direction": "LONG",
  "entry_price": 495.30,
  "stop_price": 482.09,
  "target_price": 528.33,
  "position_size": 1200,
  "leverage": 25,
  "confidence": 70.0,
  "trade_type": "SWING",
  "pending_scale_in": true
}
```

v2_closed_trades

```
{
  "_id": "trade_99",
  "symbol": "ETH/USDT",
  "pnl_pct": 1.54,
  "exit_reason": "TARGET"
}
```

trader_settings

```
{
  "_id": "v2_settings",
  "active": true,
  "min_confidence": 80,
  "position_scaling_enabled": true
}
```

10. Background Tasks

Task	Interval	Description
autonomous_trading_v2	60s	Scan markets, execute trades
free_will_v2	90s	Elite signal generation
dual_engine	120s	Day trader + Long term scans
price_alerts	30s	Check price alert triggers
trade_evaluation	30s	Evaluate trades, trail stops
self_healer	60s	Monitor service health

11. Environment Variables

Backend (.env)

Variable	Purpose
MONGO_URL	MongoDB connection string
DB_NAME	Database name (aeon_trading)
TELEGRAM_TOKEN	Telegram bot token
MEXC_API_KEY	MEXC exchange API key
EMERGENT_KEY	Emergent LLM universal key

Frontend (.env)

Variable	Value
REACT_APP_BACKEND_URL	https://aeon-paper-trade.preview.emergentagent.com

Summary

AEON is a comprehensive autonomous trading system featuring:

- ✓ 24/7 autonomous paper trading
 - ✓ 25 crypto pairs coverage
 - ✓ Weighted confluence scoring (Tech 60%, SMC 20%, Derivatives 20%)
 - ✓ Dynamic trade styles (SCALP/DAY/SWING) based on volatility
 - ✓ Position scaling (50/50 entry system)
 - ✓ Risk management (auto-blacklist, cooldowns, trailing stops)
 - ✓ AI-powered voice/text chat (GPT-4o)
 - ✓ Full web dashboard with real-time data
 - ✓ Telegram bot with 50+ commands
 - ✓ Comprehensive API (70+ endpoints)
 - ✓ MongoDB persistence
 - ✓ Self-healing system monitoring
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Document Version: 2.0

Last Updated: February 24, 2026

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